0021-972X/03/$15.00/0
Printed in U.S.A.

The Journal of Clinical Endocrinology & Metabolism 88(7):3292–3298
Copyright © 2003 by The Endocrine Society
doi: 10.1210/jc.2002-021983

The Effect of Growth Hormone on the Insulin-Like Growth Factor System during Fasting

HELENE NØRRELUND, JAN FRYSTYK, JENS OTTO LUNDE JØRGENSEN, NIELS MØLLER, JENS SANDAHL CHRISTIANSEN, HANS ØRSKOV, AND ALLAN FLYVBJERG

Medical Department M (Endocrinology and Diabetes) (H.N., J.O.L.J., N.M., J.S.C., A.F.), Aarhus Kommunehospital, and Medical Research Laboratories (J.F., N.M., H.O., A.F.), Institute of Experimental Clinical Research, Aarhus University, DK-8000 Aarhus, Denmark

The present study investigates the possible stimulatory effect of endogenous GH on IGF and IGF-binding protein (IGFBP) levels during fasting. Eight normal subjects were examined on four occasions: 1) in the basal postabsorptive state; 2) after 40 h of fasting; 3) after 40 h of fasting with somatostatin suppression of GH; and 4) after 40 h of fasting with suppression of GH and exogenous GH replacement. The two somatostatin experiments were identical in terms of hormone replacement (except for GH). Short-term fasting led to a 50% reduction in free IGF-I. The reduction in free IGF-I was paralleled by an increase in IGFBP-1, an increase in the complex formation of IGFBP-1 and IGF-I, and a modest reduction in IGFBP-3 proteolysis. GH deprivation during fasting led to a 35% reduction in total IGF-I and a 70% reduction in free IGF-I. GH replacement increased free and total IGF-I to levels similar to those observed during plain fasting and decreased IGFBP-1, however, without affecting IGFBP-1-bound IGF-I. Finally, IGFBP-3 proteolysis was slightly increased by GH replacement. In conclusion, the major new finding of the present study is that the GH hypersecretion seen during short-term fasting is not merely secondary to a reduction in IGF bioactivity. (*J Clin Endocrinol Metab* 88: 3292–3298, 2003)

FASTING CONSTITUTES A VERY robust stimulus for pituitary GH release (1, 2), and sustained nutritional deprivation causes profound changes in the IGF system. The fasting-induced decrease in circulating IGF-I has, based on data from experimental studies, been explained by decreased hepatic IGF-I mRNA expression (3, 4) and impairment of the Janus kinase-signal transducer and activator of transcription pathway (5). The turnover rate of serum IGF-I may in addition be altered by fasting.

Circulating IGF-I and -II are bound to IGF-specific binding proteins (IGFBPs). More than 99% of the circulating IGFs are bound to IGFBP-3 or contained in complexes with five other high affinity IGFBPs. The IGFBPs are believed to modulate the growth-promoting and metabolic actions of the IGFs; furthermore, the IGFBPs are modulated by specific proteases.

During nutritional deprivation, the first observed change is an increase in IGFBP-1, an *in vivo* inhibitor of IGF-I action (6). Because serum IGFBP-1 binds only a minor portion of the circulating IGF-I pool, short-term changes in IGFBP-1 do not affect the total (extractable) levels of IGF-I (7). IGFBP-1 is inversely regulated by portal insulin (8); however, GH may suppress IGFBP-1 through insulin-independent mechanisms (9). Studies based on nonfasting and overnight fasting serum samples have shown an inverse relationship between levels of IGFBP-1 and free IGF-I (7, 10), and this may serve as a mechanism to adjust IGF-I bioactivity in relation to the actual fuel supply (11). During fasting a recent study demonstrates reciprocal changes in free IGF-I and IGFBP-1 (12) and determination of the binary complex of IGF-I and IGFBP-1 (13) has confirmed that IGFBP-1 affects IGF-I bioactivity by regulating levels of free IGF-I during fasting.

Immunoreactive IGFBP-3 concentrations have previously been reported to be only slightly decreased (14) or unaltered (15) during short-term fasting. However, the IGF-binding capacity of IGFBP-3 may be altered by specific serum proteases (16–19). A role for GH in the regulation of IGFBP-3 proteolysis has been suggested by Lassarre *et al.* (20), who reported that the amount of proteolyzed IGFBP-3 was increased in GH deficiency and decreased in acromegaly, compared with normal subjects.

The aim of the present study was to investigate whether the fasting-induced increase in GH affects IGF-I levels, and if so, whether this is related to changes in circulating levels of IGFBPs or their binding capacity. We therefore compared serum levels of IGFs and IGFBPs in eight healthy subjects on four occasions (to define the specific effect of fasting and GH) in a design that allowed for control of insulin (pancreatic clamp).

Subjects and Methods

The study population comprised eight healthy males (age, 23 ± 0.4 yr; 23 ± 0.8 kg/m^2). All subjects gave their written informed consent, and the study was approved by the regional ethics committee and conducted according to the Declaration of Helsinki.

All subjects were allocated to the following studies: 1) basal postabsorptive investigations (12-h overnight fasting: basal); 2) 40 h of fasting (fast); 3) 40 h of fasting with suppression of endogenous GH secretion with somatostatin (fast-GH); and 4) 40 h of fasting with suppression of endogenous GH secretion and replacement of GH by infusion (fast+GH) (Fig. 1). During studies 3 and 4, somatostatin, insulin, and glucagon were infused for the last 28 h; in addition, GH was added to the regimen during study 4. Somatostatin (200 μg/h) was infused together with insulin [100 IE/ml Actrapid, Novo Nordisk A/S, Bagsværd, Denmark: the infusion rate being adjusted to maintain plasma glucose level between 3 and 5 mmol/liter (0.15 mU/kg·min to 0 mU/kg·min)]

Abbreviations: IGFBP, IGF-specific binding protein; IRMA, immunoradiometric assay; rh, recombinant human; tIGF, total IGF; WIB, Western immunoblotting; WLB, Western ligand blotting.

The Endocrine Society. Downloaded from press.endocrine.org by [${individualUser.displayName}] on 30 October 2015. at 18:19 For personal use only. No other uses without permission. . All rights reserved.

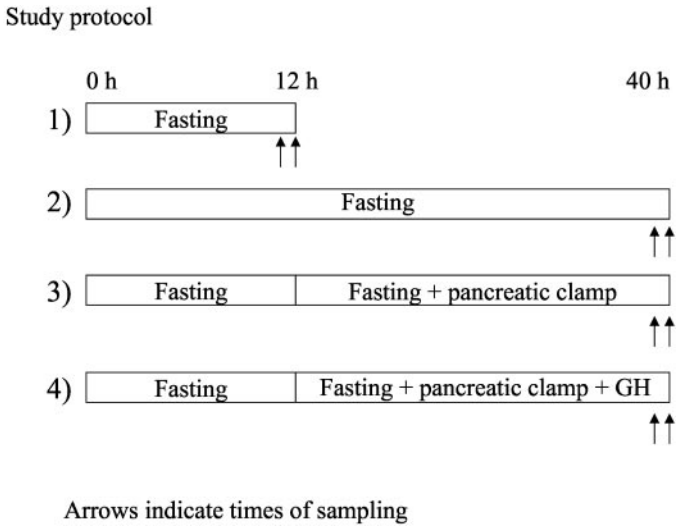

FIG. 1. Study design.

and glucagon [1 mg/ml GlucaGen, Novo Nordisk A/S; the infusion rate being adjusted to maintain glucose levels (1.0 ng/kg·min to 1.5 ng/kg·min)]. The two somatostatin experiments were identical in terms of hormone replacement doses, except for GH [12 IU/ml Norditropin, Novo Nordisk A/S; 4.5 IU given partly as bolus injections (0.38 IU every fourth hour for 24 h) and partly as continuous infusion (0.08 IU/h)]. Infusions with somatostatin, insulin, glucagon, and GH were continued for a total of 28 h. During fasting only tap water was allowed.

Assays

All samples from the same individual were analyzed within the same run, and all measurements were performed in duplicates within the same run unless otherwise stated. Total IGF-I and IGF-II were determined by noncompetitive, time-resolved immunofluorometric assays as previously described (21). Serum total (extractable) IGF (tIGF)-I and -II were determined in acid ethanol extracts and serum-free IGF-I by ultrafiltration by centrifugation as previously described (22). IGFBP-1 was determined by an in-house RIA performed as described by Westwood *et al.* (23) with modifications. In brief, breakable Maxisorb microtiter plates (Nunc, Roskilde, Denmark) were coated with a polyclonal donkey antimouse IgG (4 mg/liter, 200 μl/well; Sigma-Aldrich Corp., Copenhagen, Denmark) dissolved in sodium-carbonate buffer (pH 9.6). After an overnight incubation at 5 C, all wells were washed once using a 50 mM Tris-HCl buffer (pH 8.0) added 0.9% (wt/vol) NaCl, 0.5% (vol/vol) Tween 20, and 0.05% (wt/vol) NaN_3, and blocked with 300 μl/well phosphate buffer (40 mmol/liter, pH 8.0) added to 1% BSA (Sigma-Aldrich Corp.), 0.05% (wt/vol) NaN_3, and 0.6% (wt/vol) NaCl. After 2 h of blocking at room temperature, the wells were washed twice and added 100 μl antigen [a serial dilution of recombinant human (rh)IGFBP-1 from HyTest (Turku, Finland) or diluted serum (1 in 4)].

All antigens were dissolved in assay buffer [40 mM phosphate buffer (pH 8.0), 0.2% (wt/vol) BSA, 0.05% (wt/vol) NaN_3, 0.9% (wt/vol) NaCl, and 2% (vol/vol) Tween 20)]. In addition, 50 μl ^{125}I-labeled rh-IGFBP-1 (10,000 cpm/well) and 50 μl of a monoclonal IGFBP-1 antibody, which recognizes all human phosphoforms of IGFBP-1 (MAB 6303; Medix Biochemica, Kauniainen, Finland), were added. Both latter reagents were dissolved in assay buffer. All samples (standards, diluted serum samples, and nonspecific binding) were analyzed in duplicate. The plates were incubated for 2 d at 5 C, washed three times, and the breakable wells counted for 3 min in a gamma counter. The lower detection limit was estimated to approximately 2.5 μg/liter, the half-maximal displacement occurred at 25 μg/liter, and the upper IGFBP-1 standard was 200 μg/liter. The within and in-between assay coefficients of variation averaged less than 5% and 16%, respectively. Addition of rhIGF-I and -II (Austral Biologicals, San Ramon, CA) and rhIGFBP-2, -3, -4, and -5 (R&D Systems, Abingdon, UK) up to 10,000 μg/liter did not affect the measured concentration of IGFBP-1 to any significant degree.

A monoclonal immunofluorometric assay was used to determine the binary complex of IGF-I and total IGFBP-1 (13). Based on measurements of total IGFBP-1 and IGFBP-1-bound IGF-I, it is possible to calculate the fraction of IGFBP-1 saturated with IGF-I. Because the assay for IGFBP-1-bound IGF-I does not determine IGFBP-1-bound IGF-II, we cannot calculate the overall saturation of IGFBP-1 but only the fraction of IGFBP-1 complexed to IGF-I. IGFBP-3 was measured by immunoradiometric assay (IRMA; Diagnostic Systems Laboratories, Webster, TX) calibrated against recombinant nonglycosylated human IGFBP-3. A double monoclonal immunofluometric assay (Delfia; Wallac, Inc., Turku, Finland) was used to measure serum GH. Insulin was determined by a commercial ELISA (DAKO Corp., Glostrup, Denmark).

Western ligand blotting (WLB) and Western immunoblotting (WIB) for serum IGFBP-3

SDS-PAGE and ligand blot analyses were performed according to the method of Hossenlopp *et al.* (24). Two microliters of serum were subjected to SDS-PAGE (10% polyacrylamide) under nonreducing conditions. Samples from each subject were analyzed in the same gel. IGFBP-3 immunoblot analysis was performed with a polyclonal IGFBP-3 antibody (1:1000; Upstate Biotechnology, Inc., Lake Placid, NY) with ^{35}S-protein A (specific activity 500 Ci/mmol) (Amersham International, Amersham, Bucks, UK).

Quantification of WLB, WIB, and IGFBP-3 degradation assay

Autoradiograms of WLBs, WIBs, and the ^{125}I-IGFBP-3 degradation assay were quantified by densitometry using a CS-9001 PC dual-wavelength flying spot scanner (Shimadzu Europe, Duisburg, Germany). The relative densities of the bands were measured as arbitrary absorbance units per square millimeter.

IGFBP-3 protease assays

The IGFBP-3 protease assay was performed as previously described (25), using human recombinant [^{125}I]IGFBP-3 obtained from Diagnostic System Laboratories. [^{125}I]IGFBP-3 (~30,000 cpm) was incubated for18 h at 37 C with 2 μl serum from patients and subjected to SDS-PAGE as described above. On each gel, internal control sera from healthy, nonpregnant subjects and term pregnant women were included. Electrophoresed gels were fixed in a 7% acetic acid solution, dried, and autoradiographed. The amount of proteolysis was calculated as a ratio of the absorbance of fragmented [^{125}I]IGFBP-3 over the sum of all [^{125}I]IGFBP-3-related ODs in that lane and expressed as a percentage (*in vitro* proteolysis). In the same way, IGFBP-3 proteolysis (*in vivo* proteolysis) was calculated from WIBs as previously described (26) as a ratio of fragmented IGFBP-3 (30 and 16–18 kDa) divided by the sum of all IGFBP-3 (38–42, 30, and 16–18 kDa) in each lane.

Statistics

Results are expressed as mean ± SEM. Statistical comparisons between the study periods (basal, fast, fast-GH, and fast+GH) were analyzed by a two-way ANOVA. The *P* value resulting from this analysis is presented in *Results.* If this test was positive, the Student-Newman-Keul was employed for *post hoc* analysis. The result is presented as *asterisk* in the figures. Because part of the study was to isolate the effects of GH during fasting, data on IGFBP-1 and IGFBP-3 during fasting with and without GH were analyzed by paired *t* test as well. The results are presented in *brackets* in *Results* as well as in the figures. Data were log transformed when not normally distributed, as tested by Kolmogorov-Smirnov. A *P* value less than 0.05 was considered significant. All statistical computations were performed with SPSS for Windows, version 8.0 (SPSS, Inc., Chicago, IL).

Results

GH, IGFs, and insulin

The level of GH was significantly lower after overnight fasting and after 40 h of fasting with GH suppression, com-

The Endocrine Society. Downloaded from press.endocrine.org by [${individualUser.displayName}] on 30 October 2015. at 18:19 For personal use only. No other uses without permission. . All rights reserved.

pared with fasting alone and GH replacement situations [GH (μg/liter): 0.3 ± 0.1 (basal); 2.3 ± 0.4 (fast); 1.3 ± 0.2 (fast-GH); 2.1 ± 0.2 (fast+GH), $P < 0.01$]. Total IGF-I decreased during fasting without GH only [tIGF (μg/liter): 296 ± 30 (basal); 275 ± 32 (fast); 173 ± 22 (fast-GH); 246 ± 25 (fast+GH), $P < 0.01$], whereas free (f)IGF-I decreased during fasting and was further suppressed during GH suppression [fIGF-I (μg/liter): 0.68 ± 0.10 (basal); 0.34 ± 0.08 (fast); 0.10 ± 0.03 (fast-GH); 0.28 ± 0.09 (fast+GH), $P < 0.01$] (Fig. 2). No significant changes in IGF-II concentrations were seen during fasting, whereas suppression of GH led to a decrease [IGF-II (μg/liter): 885 ± 44 (basal); 881 ± 61 (fast); 762 ± 47 (fast-GH); 869 ± 51 (fast+GH), $P < 0.05$]. Insulin levels were decreased during fasting and somatostatin infusions, but no difference was found with or without GH substitution [insulin (pmol/liter): 25.8 ± 3.7 (basal); 11.1 ± 1.0 (fast); 5.1 ± 1.4 (fast-GH); 7.8 ± 1.2 (fast+GH), $P < 0.01$].

IGFBP-1

Forty hours of fasting caused a significant increase in IGFBP-1 ($P < 0.01$) but most pronounced during somatostatin suppression of GH [IGFBP-1 (μg/liter): 2.2 ± 0.4 (basal);

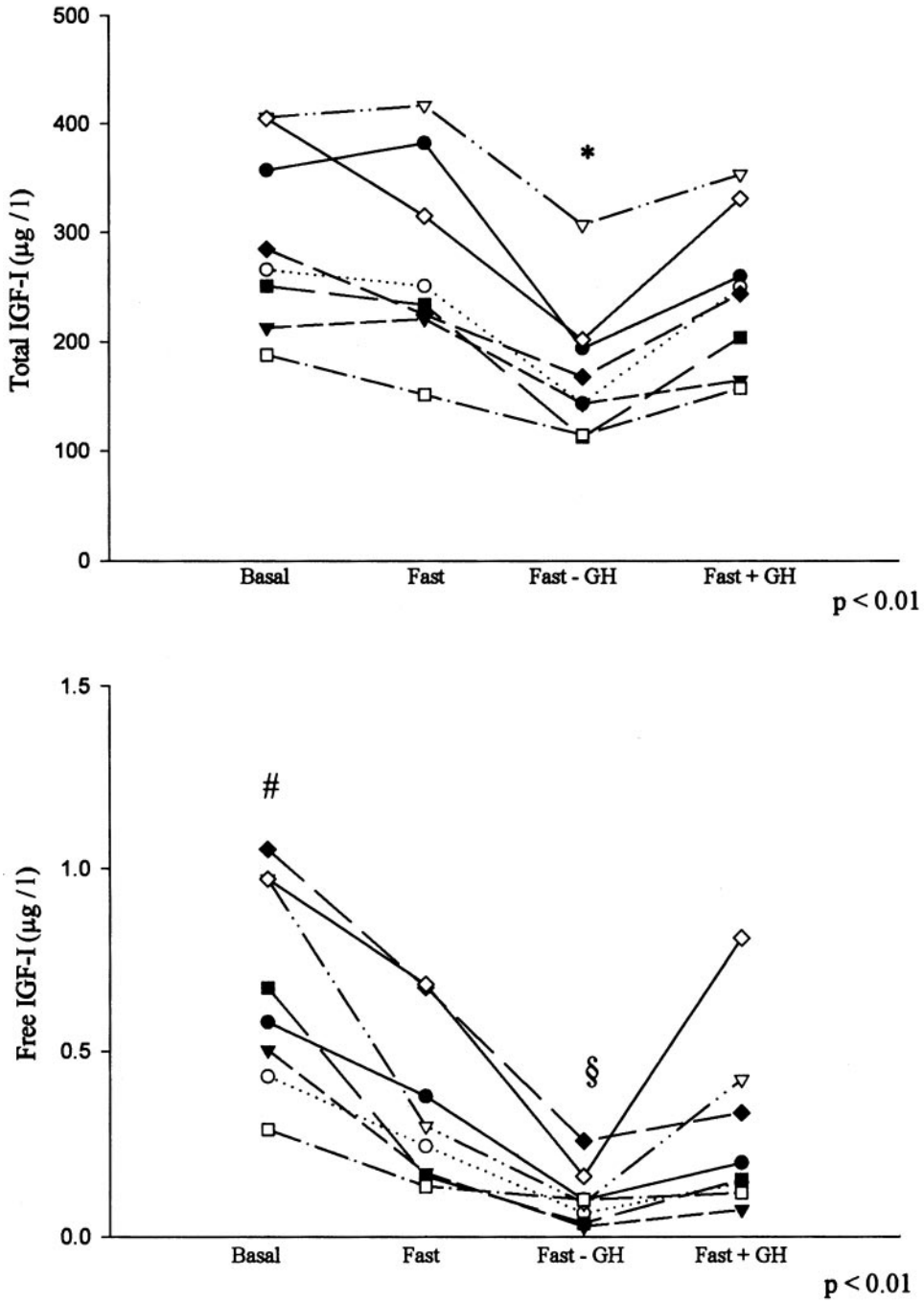

FIG. 2. Individual levels of total and free IGF-I . P value: ANOVA; *post hoc* analysis: *, $P < 0.01$ *vs.* other groups; #, $P < 0.01$ *vs.* other groups; §, $P < 0.01$ *vs.* other groups. Subject 1 (●); subject 2 (○); subject 3 (▲); subject 4 (△); subject 5 (■); subject 6 (□); subject 7 (♦); subject 8 (◇).

The Endocrine Society. Downloaded from press.endocrine.org by [${individualUser.displayName}] on 30 October 2015. at 18:19 For personal use only. No other uses without permission. . All rights reserved.

23.2 ± 3.6 (fast); 92.3 ± 8.7 (fast-GH); 37.1 ± 5.9 (fast+GH), $P < 0.01$] in spite of comparable insulin levels. Total IGFBP-1 responded similarly (Fig. 3). During fasting a parallel increase in the binary complex was observed ($P < 0.01$). The binary complex remained unchanged during fasting with and without GH (paired *t* test, $P > 0.05$), implying that the fraction of IGFBP-1 carrying IGF-I was significantly decreased during fasting without GH.

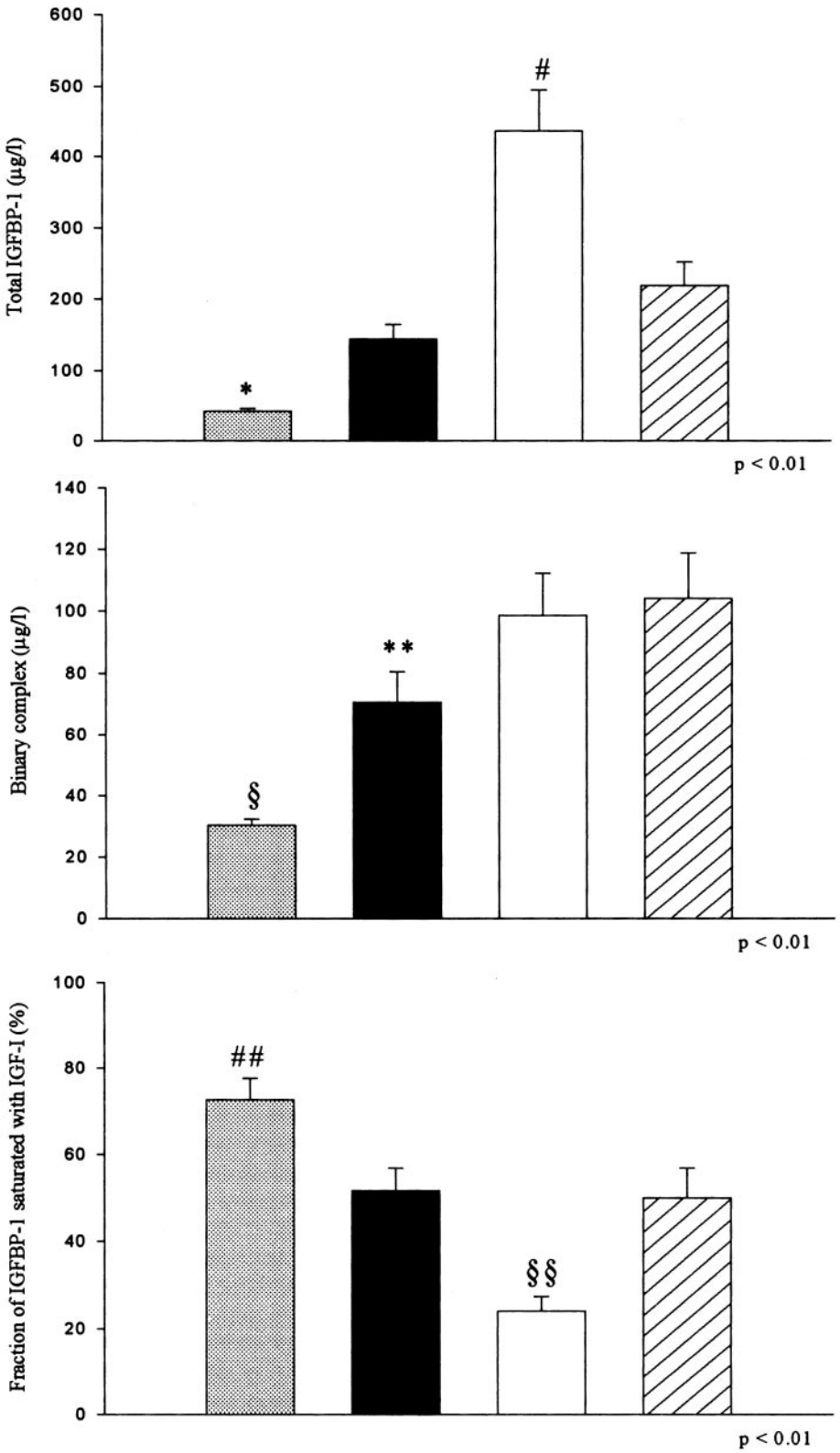

FIG. 3. Serum levels of total IGFBP-1, the binary complex, and the fraction of IGFBP-1-carrying IGF. Data are mean ± SEM. *P* value: ANOVA; *post hoc* analysis: *, $P < 0.01$ *vs.* other groups; #, $P < 0.01$ *vs.* other groups; §, $P < 0.01$ *vs.* other groups; **, $P < 0.01$ *vs.* other groups; ##, $P < 0.01$ *vs.* other groups, §§, $P < 0.01$ *vs.* other groups. Groups are shown: basal group (*hatched bar*); fast group (*black bar*); fast-GH group (*white bar*); and fast+GH group (*striped bar*).

The Endocrine Society. Downloaded from press.endocrine.org by [${individualUser.displayName}] on 30 October 2015. at 18:19 For personal use only. No other uses without permission. . All rights reserved.

IGFBP-3 concentrations determined by IRMA

A significant decrease in IGFBP-3 (Fig. 4) was observed during fasting, and a further reduction was observed during fasting without GH [IGFBP-3 (μg/liter): 3993 ± 203 (basal); 3743 ± 223 (fast); 3248 ± 184 (fast-GH); 3685 ± 171 (fast+GH), $P < 0.01$], whereas IGFBP-3 determined by WLB tended to be increased during fasting (ANOVA: $P < 0.05$; *post hoc*: $P = 0.09$). IGFBP-3 (WLB) was significantly increased during fasting with GH (paired *t* test, $P = 0.03$).

IGFBP-3 protease activity

Results are shown (Fig. 4). Levels of proteolysis did not differ when measured *in vitro* [IGFBP-3 proteolysis (%): 21 ± 1 (basal); 21 ± 1 (fast); 21 ± 1 (fast-GH); 22 ± 2 (fast+GH), $P > 0.05$], whereas IGFBP-3 proteolysis measured *in vivo* was significantly decreased after 40 h of fasting, compared with proteolysis after fasting overnight [IGFBP-3 proteolysis (%): 40.8 ± 2.4 (basal); 35.6 ± 2.2 (fast); 38.4 ± 1.4 (fast-GH); 40.7 ± 1.6 (fast+GH), $P < 0.05$]. The *in vivo* IGFBP-3 proteolysis was significantly increased during fasting with GH (paired *t* test, $P = 0.04$).

Discussion

Fasting causes a state of GH resistance characterized by GH hypersecretion and low serum IGF levels. The present study was designed to investigate the effect of the fasting-induced GH hypersecretion on the circulating IGF system. We found that in spite of unchanged total IGF-I levels, plain short-term fasting led to a 50% reduction in free IGF-I. The reduction in free IGF-I was accompanied by an increase in IGFBP-1 and IGFBP-1-complexed IGF-I, and a modest reduction in IGFBP-3 proteolysis. Deprivation of GH during short-term fasting led to further reductions in total and free IGF-I and elevations in IGFBP-1; changes that were all fully abolished by GH replacement. Finally, GH replacement caused a slight increase in IGFBP-3 proteolysis.

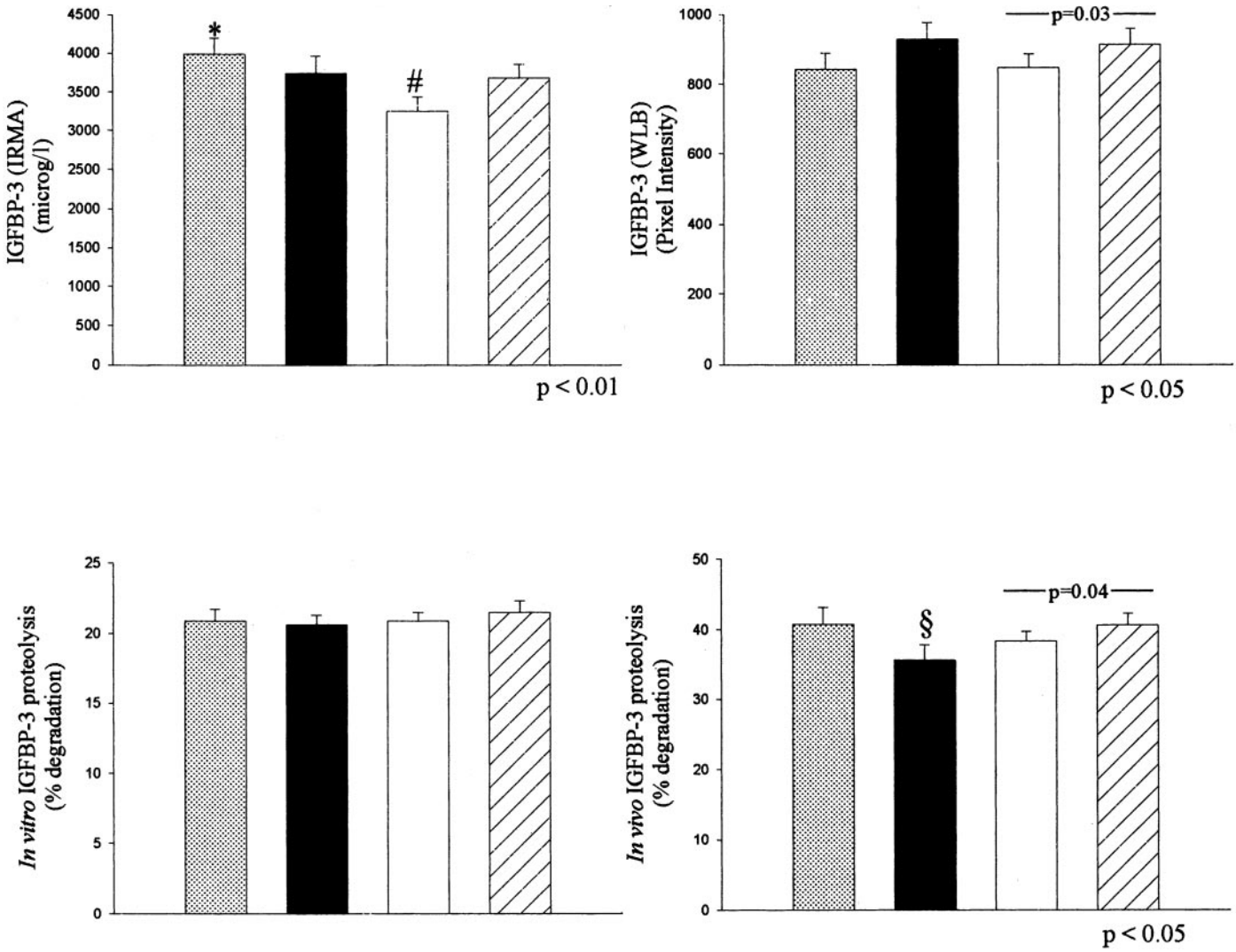

FIG. 4. Determination of IGFBP-3 by IRMA and WLB. IGFBP-3 proteolysis *in vitro* and *in vivo*. All data are mean ± SEM. *P* value: ANOVA; *post hoc* analyses: *, $P < 0.01$ *vs.* other groups; #, $P < 0.01$ *vs.* other groups; §, $P < 0.05$ *vs.* basal. Paired samples *t* test (fast-GH *vs.* fast+GH): $P = 0.03$; $P = 0.04$. Groups are shown: basal group (*hatched bar*); fast group (*black bar*); fast-GH group (*white bar*); and fast+GH group (*striped bar*).

The Endocrine Society. Downloaded from press.endocrine.org by [${individualUser.displayName}] on 30 October 2015. at 18:19 For personal use only. No other uses without permission. . All rights reserved.

IGFBP-1 is considered to be an important regulator of the bioactive fraction of the circulating IGF-I pool (8), and low circulating levels are usually associated with conditions of increased IGF-I activity (*e.g.* the postprandial phase and obesity) (8, 22, 27). An increase in IGFBP-1 during prolonged fasting in healthy adults has been demonstrated earlier (28), and short-term fasting in humans and rats has shown an inverse correlation between IGFBP-1 and free IGF-I (12, 29). Previous studies have been based on correlation studies. However, by use of an assay, which allows determination of the binary complex of IGFBP-1 bound IGF-I (13), the decrease in free IGF-I during fasting can be shown to be explained by increased complex formation with IGFBP-1. Considering the anabolic and hypoglycemic actions of IGF-I, the increase in IGFBP-1 during fasting (28, 30, 31) may be regarded as a protective mechanism to avoid hypoglycemia.

The undisputed close relationship between insulin and IGFBP-1 (8) makes it difficult to scrutinize putative direct GH effects. Observations in rats made Murphy *et al.* (32) suggest that GH deficiency rather than insulin deficiency is responsible for the fasting-induced increase in IGFBP-1. During conditions of low, suppressed, or declining insulin levels, a suppressive effect of GH on IGFBP-1 level also appears to be unmasked in humans (9). The present study showed that fasting increased IGFBP-1 as well as the complex formation of IGF-I and IGFBP-1. Thus, the reduction in free IGF-I during fasting is likely to be caused by an up-regulation of IGFBP-1, as previously reported (13). When GH was suppressed by somatostatin infusion, fasting caused a further increase in IGFBP-1. The up-regulation in serum IGFBP-1 is likely to be explained by the lack of GH (9), because GH replacement restored IGFBP-1 to the plain fasting level. Noteworthy, the further increase in IGFBP-1 caused by GH suppression did not result in a concomitant increase in IGFBP-1/IGF-I binary complex formation. Intuitively, this would have been expected, and it therefore indicates that there was no more IGF-I available for complex formation with IGFBP-1. We have previously shown that during 3 d of fasting in rats, which causes a marked suppression of the circulating IGF-system, changes in free IGF-I appears to be more closely related to total IGF-I than IGFBP-1 (29). Our present study indicates that the same may be true in humans (*i.e.* as long as total IGF-I remains relatively stable as seen during plain fasting, changes in free IGF-I are closely related to IGFBP-1). However, when total IGF-I undergoes marked changes as seen during GH suppression, then total IGF-I has a larger impact on free IGF-I than IGFBP-1.

Our study showed that GH hypersecretion during fasting, despite nutritional deprivation, was able to maintain total (and free) IGF-I levels at a reasonable level, which was clearly shown during GH suppression. These changes can be explained by a maintained *de novo* synthesis of IGF-I. Previous studies demonstrating a reduction in hepatic IGF-I production (3, 4, 33) and impairment of the Janus kinase-signal transducer and activator of transcription signaling pathway (5) during fasting have been performed in rats. In rats, unlike in humans, the secretion of GH becomes markedly suppressed by fasting (11, 29, 34), and this makes such data difficult to extrapolate to humans.

The physiological significance of IGFBP-3 proteolysis remains unknown, but the decreased ligand affinity of the fragments may increase IGF availability to tissues and cells and thereby increase IGF bioactivity (19). Numerous conditions have been described in which IGFBP-3 proteolysis is induced, many of them characterized by an acute or chronic catabolic status (16, 18, 35–37). In the present study, IGFBP-3 *in vivo* proteolysis decreased modestly during fasting. This could increase IGFBP-3 affinity for IGF and lead to a further reduction in free IGF-I. When measuring IGFBP-3 by WLB analysis, which does not detect IGFBP-3 fragments (19), we found nearly significant increase in IGFBP-3 during plain fasting ($P = 0.09$), suggesting a concomitant increase in intact high-affinity IGFBP-3.

About 80% of circulating IGFs are complexed with IGFBP-3 and acid-labile subunit, forming a high-molecular-mass (150 kDa) ternary complex (38). GH is the principal regulator of all three components of the 150-kDa complex. Furthermore, a possible role for GH in the regulation of IGFBP-3 proteolysis has been suggested by Lassarre *et al.* (20), who observed that the relative amount of proteolyzed IGFBP-3 on WIB was increased in GH-deficient patients and decreased in untreated acromegaly. In a more recent study from the same group (39), the percentage of proteolyzed IGFBP-3 was within the normal range in GH-deficient patients but elevated in acromegalics. When comparing fasting with and without GH, GH suppression was associated with a decrease in IGFBP-3 measured by IRMA and a decrease in intact IGFBP-3 (WLB). As opposed to the finding by Lassarre *et al.* (20, 39), this was associated with a decrease in IGFBP-3 proteolysis *in vivo,* suggesting an increased IGFBP-3 affinity for IGF, which both may contribute to the reduction in IGF-I. GH replacement caused an increase in IGFBP-3 proteolysis. The reason for this discrepancy is obscure but may be caused by differences in the antibodies used. The IRMA used for measurements of IGFBP-3 in the present study detects intact IGFBP-3 as well as fragments of IGFBP-3. WLB, on the other hand, exclusively detects intact IGFBP-3. Other studies, using the same antibodies as we used, have shown an increased *in vivo* proteolysis during GH administration to healthy subjects (26) but failed to show any effect of GH on the degree of IGFBP-3 proteolysis in GH-deficient patients or after major surgery (26, 40).

In conclusion, our data show that somatostatin-induced suppression of GH during fasting leads to a further reduction in circulating IGF-I than caused by fasting *per se,* whereas substitution of GH added in physiological doses to somatostatin-treated subjects, is able to restore serum-free and total IGF-I to levels similar to what was seen during plain fasting. These data provide evidence that the fasting induced increase in GH is not merely secondary to a reduced IGF-I activity but in itself has an important stimulatory effect on circulating IGF-I levels. This study does not allow us to make any solid conclusion on the mechanisms by which GH stimulated the IGF-system during fasting, but GH may act through two different pathways: first, GH reduces IGFBP-1, which is likely to increase free IGF-I. Second, GH may directly stimulate IGF-I synthesis, which is important for free as well as total IGF-I. GH also appeared to increase IGFBP-3 proteolysis during fasting, and this may add to the effects of GH on the activity of the IGF-system during fasting.

The Endocrine Society. Downloaded from press.endocrine.org by [${individualUser.displayName}] on 30 October 2015. at 18:19 For personal use only. No other uses without permission. . All rights reserved.

Acknowledgments

We are grateful to Karen Mathiasen, Eva Seier Petersen, and Lone Korsgård for excellent technical assistance.

Received December 17, 2002. Accepted April 1, 2003.

Address all correspondence and requests for reprints to: Helene Nørrelund, M.D., Ph.D., Medical Department M, Aarhus Kommunehospital, DK-8000 Aarhus C, Denmark. E-mail: helenenorrelund@dadlnet.dk.

This work was supported by the Danish Research Council Grant 9600822 (Aarhus University–Novo Nordisk Centre for Research in Growth and Regeneration) and Grants 9700592 and 22020141; the Eva and Henry Frænkels Memorial Foundation; and the Hørslev Foundation.

References

1. **Ho KY, Veldhuis JD, Johnson ML, Furlanetto R, Evans WS, Alberti KGMM, Thorner MO** 1988 Fasting enhances growth hormone secretion and amplifies the complex rhythms of growth hormone secretion in man. J Clin Invest 81:968–975
2. **Hartman ML, Veldhuis JD, Johnson ML, Johnson ML, Lee MM, Alberti KGMM, Samojlik E, Thorner MO** 1992 Augmented growth hormone (GH) secretory burst frequency and amplitude mediate enhanced GH secretion during a two-day fast in normal men. J Clin Endocrinol Metab 74:757–765
3. **Elmer CA, Schalch DS** 1987 Nutritional-induced changes in hepatic insulin-like growth factor I (IGF-I) gene expression in rats. Endocrinology 120:824–832
4. **Lowe WL, Adamo M, Werner H, Roberts CT, LeRoith D** 1989 Regulation by fasting of rat insulin-like growth factor I and its receptor. Effects on gene expression and binding. J Clin Invest 84:619–626
5. **Beauloye V, Willems B, de Coninck V, Frank SJ, Edery M, Thissen JP** 2002 Impairment of liver GH receptor signaling by fasting. Endocrinology 143: 792–800
6. **Cox GN, McDermott MJ, Merkel E, Stroh CA, Ko SC, Squires CH, Gleason TM, Russell D** 1994 Recombinant human insulin-like growth factor (IGF)-binding protein-1 inhibits somatic growth stimulated by IGF-I and growth hormone in hypophysectomized rats. Endocrinology 135:1913–1920
7. **Frystyk J, Grøfte T, Skjærbæk C, Ørskov H** 1997 The effect of oral glucose on serum free insulin-like growth factor-I and -II in healthy adults. J Clin Endocrinol Metab 82:3124–3127
8. **Lee PD, Conover CA, Powell DR** 1993 Regulation and function of insulin-like growth factor-binding protein-1. Proc Soc Exp Biol Med 204:4–29
9. **Nørrelund H, Fisker S, Vahl N, Børglum J, Richelsen B, Christiansen JS, Jørgensen JO** 1999 Evidence supporting a direct suppressive effect of growth hormone on serum IGFBP-1 levels. Experimental studies in normal, obese and GH-deficient adults. Growth Horm IGF Res 9:52–60
10. **Frystyk J, Grønbæk H, Skjærbæk C, Flyvbjerg A, Ørskov H, Baxter RC** 1998 Developmental changes in serum levels of free and total insulin-like growth factor-I (IGF-I), IGF-binding protein-1 and -3 and the acid labile subunit in rats. Endocrinology 139:4286–4292
11. **Thissen JP, Ketelslegers J-M, Underwood LE** 1994 Nutritional regulation of the insulin-like growth factors. Endocr Rev 15:80–101
12. **Katz LEL, DeLeon DD, Zhao H, Jawad AF** 2002 Free and total insulin-like growth factor (IGF)-I levels decline during fasting: relationships with insulin and IGF-binding protein-1. J Clin Endocrinol Metab 87:2978–2983
13. **Frystyk J, Højlund K, Rasmussen KN, Jørgensen SP, Wildner-Christensen M, Ørskov H** 2002 Development and clinical evaluation of a novel immunoassay for the binary complex of IGF-I and IGF-binding protein-1 in human serum. J Clin Endocrinol Metab 87:260–266
14. **Blum WF, Ranke MB** 1990 Plasma IGFBP-3 levels as clinical indicators. In: Spencer EM, ed. Modern concepts of insulin-like growth factors. Amsterdam: Elsevier; 381–393
15. **Bang P, Brismar K, Rosenfeld RG, Hall K** 1994 Fasting affects serum insulin-like growth factors (IGFs) and IGF-binding proteins differently in patients with noninsulin-dependent diabetes mellitus versus healthy nonobese and obese subjects. J Clin Endocrinol Metab 78:960–967
16. **Giudice LC, Farrell EM, Pham H, Lamson G, Rosenfeld RG** 1990 Insulin-like growth factor binding proteins in maternal serum throughout gestation and in the puerperium: effects of a pregnancy-associated serum protease activity. J Clin Endocrinol Metab 71:806–816
17. **Davenport MS, Isley WL, Pucilowska JB** 1992 Insulin-like growth factor-binding protein-3 proteolysis is induced after elective surgery. J Clin Endocrinol Metab 75:590–595
18. **Cwyfan Hughes SC, Cotteril AW, Molloy AR** 1992 The induction of specific proteases for insulin-like growth factor-binding proteins following major heart surgery. J Endocrinol 135:135–145
19. **Suikkari A-M, Baxter RC** 1991 Insulin-like growth factor (IGF) binding protein-3 in pregnancy serum binds native IGF-I but not iodo-IGF-I. J Clin Endocrinol Metab 73:1377–1379
20. **Lassarre C, Lalou C, Perin L, Binoux M** 1994 Protease-induced alteration of insulin-like growth factor binding protein-3 as detected by radioimmunoassay. Agreement with ligand blotting data. Growth Regul 4:48–55
21. **Frystyk J, Dinesen B, Ørskov H** 1995 Non-competitive time-resolved immunofluorometric assays for determination of human insulin-like growth factor I and II. Growth Regul 5:169–176
22. **Frystyk J, Skjærbæk C, Dinnesen B, Ørskov H** 1994 Free insulin-like growth factors (IGF-I and IGF-II) in human serum. FEBS Lett 348:185–191
23. **Westwood M, Gibson JM, Davies AJ, Young RJ, White A** 1994 The phosphorylation pattern of insulin-like growth factor-binding protein-1 in normal plasma is different from that in amniotic fluid and changes during pregnancy. J Clin Endocrinol Metab 79:1735–1741
24. **Hossenlopp S, Seurin D, Segovia-Quinson B, Hardouin S, Binoux M** 1986 Analysis of serum insulin-like growth factor binding proteins using Western blotting: use of the method for titration of the binding proteins and competitive binding studies. Anal Biochem 154:138–143
25. **Lamson G, Giudice LC, Rosenfeld RG** 1991 A simple assay for proteolysis of IGFBP-3. J Clin Endocrinol Metab 72:1391–1393
26. **Skjærbæk C, Kaal A, Møller J, Vahl N, Weeke J, Ørskov H, Flyvbjerg A** 1998 No effect of growth hormone on serum insulin-like growth factor binding protein-3 proteolysis. J Clin Endocrinol Metab 83:1206–1210
27. **Conover CA, Lee PD, Kanaley JA, Clarkson JT, Jensen MD** 1992 Insulin regulation of insulin-like growth factor binding protein-1 in obese and nonobese humans. J Clin Endocrinol Metab 74:1355–1360
28. **Cotterill AM, Holly JM, Wass JA** 1993 The regulation of insulin-like growth factor binding protein (IGFBP)-1 during prolonged fasting. Clin Endocrinol Oxf 39:357–362
29. **Frystyk J, Delhanty PJD, Skjærbæk C, Baxter RC** 1999 Changes in the circulating IGF system during short-term fasting and refeeding in rats. Am J Physiol 277:E245–E252
30. **Lewitt MS, Saunders H, Phyual JL, Baxter RC** 1994 Regulation of insulin-like growth factor-binding protein-1 in rat serum. Diabetes 43:232–239
31. **Smith WJ, Underwood LE, Clemmons DR** 1995 Effects of caloric or protein restriction on insulin-like growth factor-I (IGF-I) and IGF-binding proteins in children and adults. J Clin Endocrinol Metab 80:443–449
32. **Murphy LJ, Seneviratne P, Moreira P, Reid RE** 1991 Enhanced expression of insulin-like growth factor-binding protein-1 in the fasted rat: the effects of insulin and growth hormone administration. Endocrinology 128:689–696
33. **Maes M, Underwood LE, Ketelslegers J-M** 1983 Plasma somatomedin-C in fasted and refed rats: close relationship with changes the liver somatogenic but not lactogenic binding sites. J Endocrinology 97:243–252
34. **Tannenbaum GS, Rorstad O, Brazeau P** 1979 Effects of prolonged food deprivation on the ultradian growth hormone rhythm and immunoreactive somatostatin tissue levels in the rat. Endocrinology 104:1733–1738
35. **Timmins AC, Cotteril AW, Cwyfan Hughes SC** 1996 Critical illness is associated with low circulating concentrations of insulin-like growth factors-I and -II, alterations in insulin-like growth factor binding proteins, and induction of an insulin-like growth factor binding protein 3 protease. Crit Care Med 24: 1460–1466
36. **Bereket A, Lang CH, Blethen SL, Fan J, Frost RA, Wilson TA** 1995 Insulin-like growth factor binding protein-3 proteolysis in children with insulin-dependent diabetes mellitus: a possible role for insulin in the regulation of IGFBP-3 protease activity. J Clin Endocrinol Metab 80:2282–2288
37. **Lalou C, Binoux M** 1993 Evidence that limited proteolysis of insulin-like growth factor binding protein-3 (IGFBP-3) occurs in the normal state outside of the bloodstream. Regul Pept 48:179–188
38. **Jones JI, Clemmons DR** 1995 Insulin-like growth factors, and their binding proteins: biological actions. Endocr Rev 16:3–34
39. **Lassarre C, Duron F, Binoux M** 2001 Use of the ligand immunofunctional assay for human insulin-like growth factor (IGF) binding protein-3 (IGFBP-3) to analyze IGFBP-3 proteolysis and IGF-I bioavailability in healthy adults, GH-deficient and acromegalic patients, and diabetics. J Clin Endocrinol Metab 86:1942–1952
40. **Skjærbæk C, Frystyk J, Ørskov H, Kissmeyer-Nielsen P, Jensen MB, Laurberg S, Møller N, Flyvbjerg A** 1998 Differential changes in free and total insulin-like growth factor I after major, elective abdominal surgery: the possible role of insulin-like growth factor-binding protein-3 proteolysis. J Clin Endocrinol Metab 83:2445–2449

The Endocrine Society. Downloaded from press.endocrine.org by [${individualUser.displayName}] on 30 October 2015. at 18:19 For personal use only. No other uses without permission. . All rights reserved.